

Evan Mullins

evangmullins@gmail.com • 314-760-8042 • <https://www.linkedin.com/in/evan-mullins> • <https://github.com/emullins16>

EDUCATION

Bachelor of Science in Materials Science and Engineering

Expected May 2029

University of Florida, Gainesville, FL

GPA: 4.00/4.00

- Relevant Coursework: Junior Materials Lab, Error Analysis & Optimization Methodologies, Mechanics of Materials, Thermodynamics of Materials, Introduction to Inorganic Materials

EXPERIENCE

Carbon Fiber Product Development Intern – Zoltek Corporation

May 2026 – August 2026

- Defined conductivity thresholds for CFRTP composites to inform sales team technical claims by designing and executing 400 trials at 8 fiber loading levels across 4 percolation studies per ASTM, ISO, and IEC standards.
- Redesigned the fiber sizing verification process by engineering a DCM extraction and recovery system with an integrated cooling chamber, reducing processing time by 75% and doubling batch capacity.
- Characterized thermal and compositional properties for new product development, historical test verification, and NCR compliance across 20 pilot batches using DMA, TGA, FTIR, and DSC.
- Optimized catalyst type and loading across 20 candidates using gel, viscosity, and exotherm testing across 6 pilot resin systems, establishing pot life benchmarks adopted during pilot cycles.

Composites Development Engineer – Swamp Launch, University of Florida

April 2026 – Present

- Constructed and designed CF/fiberglass filament winder for rapidly prototyped composite airframes and COPVs in alignment with Swamp Launch's 5-year Spaceshot Development team.
- Reduced winder time by 50%, validated across 5 test runs, by rebuilding Python software to automate G-code generation.
- Developed ASTM-based lap shear test protocol for structural foam/prepreg bonding across 2 systems to optimize fin core compatibility and composite quality.

IREC Composites Engineer – Swamp Launch, University of Florida

August 2025 – Present

- Manufactured 10+ SRAD composite parts per cycle (airframes, fins, couplers, nosecone), controlling epoxy formulation, vacuum cure, and post-processing to reduce defects and decrease layup time by 50%.
- Reduced competition fin thickness by 48% while maintaining 80% structural integrity by executing the team's first tip-to-tip layup trial and prepreg/autoclave sandwich panel cure.

PROJECTS

HAB Payload and Thermal Engineer – NASA Florida Space Grant Consortium, KSC Visitor Complex

July 2026 – Present

- Designed and optimized insulation material selection and payload design to analyze convective heat transfer collapse of aluminum components during 100,000 ft flight, validated via Python thermal model against operating temperature constraints.
- Optimized BOM across 3 payloads within an \$800 budget, coordinating lead times to greenlight all experiments for flight.
- Selected by FLSGC to produce audiovisual documentation for content briefed to Florida's congressional delegation, Space Trek websites, and social media of 2,000 followers.

Wildfire-Resistant Composite UAV Radome – EMA 3011 Design Project

April 2026

- Recommended 30% GF-PEEK as optimal VTOL radome material, achieving > 100°C higher service temperature than baseline PU foams while meeting RF transparency, manufacturing, and strength constraints using a multi-constraint merit index to evaluate candidate materials in Granta EduPack.

LEADERSHIP

Outreach Director – Freshman Leadership in Engineering, University of Florida

April 2026 – Present

- Oversaw 10 freshmen and organized monthly projects averaging 20 attendees, streamlining logistics, communication, and finances as Outreach Director for a program selecting from 800+ applicants.
- Led "Pack it Pink" fundraiser for breast cancer patients, generating \$2,613 in 12 days (program record).

SKILLS

Manufacturing & Lab: DOE, ANOVA, Regression Modeling, Autoclave Curing, RTM, Vacuum Infusion, NDT (Ultrasonic), Instron Testing, TGA, DMA, DSC, FTIR, 5S Lean Principles, 3D Printing, Milling, Failure Analysis

Engineering Software: Fusion360, Ansys Granta, OpenRocket, JMP, ImageJ, OpenSim

Languages: Python (sklearn, scipy, numpy, matplotlib, pandas), G-code, MATLAB, HTML, JavaScript

Interests: Jazz Saxophone, Filmmaking, Free Verse Poetry, Volleyball, Foreign Languages